# Where are you? How far did you travel?

**LEARNING TARGET** I can describe a route using an origin, signed position, total distance, and displacement.

## A · KEY VOCABULARY

| | |
|---|---|
| **Position** | A signed coordinate in a reference frame: choose an origin and a + direction. $x$ = −2 m is 2 m left of the origin. |
| **Distance** | Sum of the length of every leg: $\lvert x_2 - x_1\rvert + \lvert x_3 - x_2\rvert + \ldots$ Never negative. |
| **Displacement** | $\Delta x = x_f - x_i$. Its sign depends on the axis you chose; the physical motion does not. |
| **Scalar / vector** | A scalar has magnitude + units (distance, time). A vector also has direction (displacement). |

## B · KEY EQUATIONS + UNITS

$$\Delta x = x_f - x_i$$

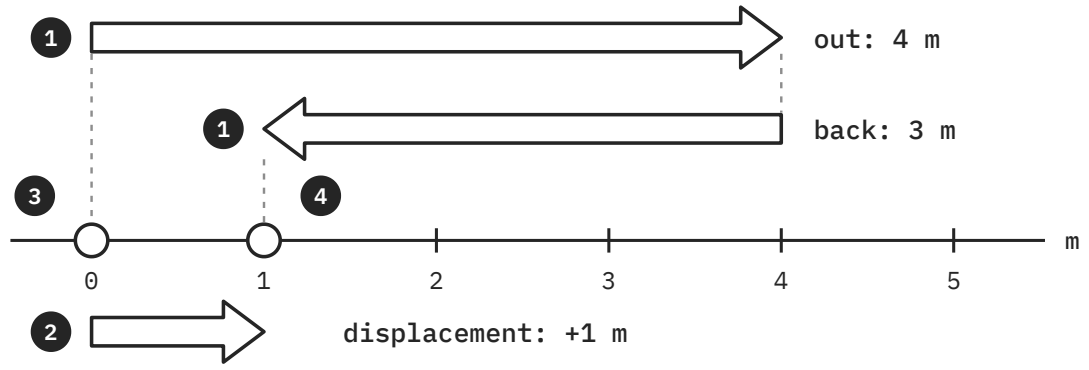
$$\text{distance} = \Sigma\ \lvert\text{each leg}\rvert$$

$\Delta$ "change in" (final − initial)

$x$ signed position, meters (m)

## C · COLOR IT · ONE TRIP, THREE QUESTIONS

*0 → 4 → 1 m: $\lvert 4 - 0\rvert + \lvert 1 - 4\rvert = 7$ m; $\Delta x = +1$ m*




- ☐ **1 · PENCIL** Distance: trace both legs
- ☐ **2 · BLUE** Displacement arrow
- ☐ **3 · BLUE** Start position
- ☐ **4 · BLUE** Finish position

## D · WORKED EXAMPLE

Start at −3 m, move to +2 m, then finish at −1 m. Find the distance and the displacement, with units.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $x_i$ = −3 m<br>turn at +2 m<br>$x_f$ = −1 m | distance = $\Sigma\ \lvert\text{leg}\rvert$<br>$\Delta x = x_f - x_i$ | $\lvert 2 - (-3)\rvert + \lvert -1 - 2\rvert = 5$<br>+ 3<br>−1 − (−3) | distance **8 m**<br>displacement **+2 m** |

## E · THE 2026-XJ THREAD

DAY 2 OF 22 · NEW TOOL: A REFERENCE FRAME

XJ's position only means something in a reference frame: an origin, axes, and a sign convention. Its displacement, not its path length, decides where it ends up.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else